a) Brand or Product Name
Mega Mount Medicinal Oxygen 100% v/v.

b) Name and Strength of Active Substance(s)
Medicinal Oxygen 100%v/v, medicinal gas, compressed.
There are no excipients.

c) Product Description
Medicinal oxygen gas is colourless and odorless.
The product is stored in cylinders with a black body and white shoulder.
It is compressed gas up to 200 bar at 27 Degree Celsius.
Medicinal oxygen contains not less than 99.5%v/v of oxygen.

d) **Pharmacodynamics**
Pharmacotherapeutic Group- Medical Gas
ATC Code: V03AN1

Normobaric oxygen
Oxygen is present in the atmosphere at 21%. Oxygen is essential for all living organisms, and continuous oxygenation of tissues is necessary to support cellular energy production. When oxygen is inhaled, it enters the lungs, diffuses through the walls of the alveoli and nearby blood capillaries, and then enters the bloodstream, where it is primarily carried by hemoglobin to the rest of the body. This process is crucial for the body's survival. Administering extra oxygen to patients experiencing hypoxia enhances the oxygen supply to their tissues.

Hyperbaric oxygen
Hyperbaric oxygen therapy (HBOT) involves pressurized oxygen, which greatly increases the amount of oxygen absorbed into the blood, including the portion not bound to hemoglobin. This process enhances the oxygen supply to body tissues. In treating gas or air embolisms, high-pressure hyperbaric oxygenation reduces the volume of gas bubbles, allowing the gas to be more effectively absorbed into the bloodstream and subsequently expelled from the lungs through exhalation.

e) Pharmacokinetics
Inhaled oxygen is absorbed through a pressure-dependent gas exchange between the alveoli and the capillary blood that flows past them. The oxygen, mostly bound to hemoglobin, is then carried through the systemic circulation to all body tissues, with only a small fraction being dissolved freely in the plasma. Oxygen plays a crucial role in cellular energy production, specifically in the aerobic production of ATP within the mitochondria. Nearly all of the oxygen absorbed by the body is eventually exhaled as carbon dioxide, produced during this metabolic process.

Absorption
Medicinal Oxygen typically administered via inhalation. It is absorbed directly into the bloodstream through the alveoli in the lungs. Oxygen diffuses across the alveolar membrane into the pulmonary capillaries based on a concentration gradient (from higher to lower partial pressure). Oxygen is bound to the haemoglobin and also dissolved in plasma.

Distribution
Medicinal oxygen is distributed by the systemic circulation and mainly transported by the haemogoblin (oxyhaemogoblin). Oxygen is distributed throughout the body and delivered to tissues based on the metabolic needs of different organs. It diffuses from the blood into tissues where it is needed for cellular respiration.

Biotransformation
Oxygen plays a crucial role in cellular metabolism, particularly in the mitochondria, where it is used in the production of adenosine trisphosphate ATP) through oxidative phosphorylation.

Elimination
While oxygen itself is not excreted, the metabolic processes it supports produce carbon dioxide (CO_2) as a byproduct.CO_2 is then transported back to the lungs, where it is exhaled.

f) Indication

Normobaric oxygen is used for:
i) Treatment or prevention of acute or chronic hypoxemia, irrespective of genesis.
ii) As the driving gas in nebulisation therapy.
iii) As part of the fresh gas supply in anaesthesia or intensive care.
iv) As first aid treatment with 100% oxygen in decompression accidents, and suspected carbon monoxide poisoning. The treatment is indicated in all age groups.
v)Treatment of an acute attack in patients with an established diagnosis of cluster headache.
This treatment is indicated in adults only.

Hyperbaric oxygen is used for:
Medicinal oxygen under hyperbaric pressure (HBO) is used for treatment of conditions where it is beneficial to increase the oxygen content of blood and other tissues above which is attainable under normobaric pressure.
i) Treatment of decompression sickness, air/gas emboli of other genesis
ii) As adjunctive treatment for osteo-radionecrosis and clostridial myonecrosis (gas gangrene).
iii) In carbon monoxide poisoning. HBO therapy is indicated essentially in patients who are or have been unconscious, that have shown neurological signs, cardiovascular dysfunction or severe acidosis and in pregnant females all irrespective of carbon monoxide haemoglobin (COHb) levels.
The treatment can be used in all age groups.

g) Recommended Dosage
Normobaric oxygen
General recommendations
The main goal of oxygen therapy, that is correction of hypoxia, is to ensure that the partial arterial oxygen pressure(PaO) is maintained at no less than 8.0 kPa (60 mmHg) or that the oxygen saturation of haemoglobin in arterial blood(SaO) is not less than 90%. This is accomplished by adjusting the fraction of oxygen in inhaled gas. The lowest oxygen fraction of inhaled gas needed to achieve the desired result of therapy i.e. a safe PaO should be used. The therapy should be evaluated continuously, and the effect of treatment measured with PaO/SaO or an estimate of SaO i.e.SpO. The oxygen fraction of inspired gas should be adjusted according to each individual patient's unique requirement, taking the risk of oxygen toxicity into account (See Symptoms and Treatment of Overdose). In cases of severe hypoxia, oxygen fractions that carry a risk of oxygen toxicity may still be warranted.

Acute or chronic hypoxia - Spontaneous breathing - Short term therapy
In emergency medicine, oxygen is typically administered via nasal prongs with at flowrate of 2 to 6 l/min or through a face mask with a flow rate of 5 to 10 l/min. For patients not at risk of respiratory failure and with an initial SpO <85%, treatment can involve using a mask with a reservoir at 10–15 L/min.
Patients with known suspected chronic respiratory disease (e.g. COPD) that may have reduced chemoreceptor sensitivity should be treated with caution as a too liberal use of oxygen may cause respiratory depression. When 100%oxygen is necessary, a face mask with a reservoir should be used, ensuring the oxygen flow is sufficient to keep the reservoir partially or fully inflated (i.e., not collapsed during breathing) or a demand valve system.
The fraction of oxygen in the inspired gas should be maintained to ensure that, with or without Positive End-Expiratory Pressure (PEEP) or Continuous Positive Airway Pressure (CPAP), a PaO >8 kPa is achieved. The effectiveness of short-term oxygen therapy should be monitored through repeated measurements of PaO or pulse oximetry, which provides a numerical value for SpO2. However, these measurements are indirect indicators of tissue oxygenation, making clinical assessment of the treatment critically important.

Acute or chronic hypoxia - Spontaneous breathing - Long-term therapy
In long-term therapy, oxygen can be administered using specially designed masks, such as Venturi masks, which allow for adjustable oxygen concentrations based on the gas flow and the mask's valve. Concentrations ranging from 24% to35% are commonly used.
The need for medicinal oxygen should be determined by obtaining arterial blood gas values and/or by monitoring SpO. The inspired oxygen should be titrated when used for long term oxygen therapy in patients with chronic hypoxic respiratory failure. A SaO/SpO between 88 and 92% is commonly assessed as adequate in patients with chronic obstructive pulmonary disease (COPD). A too liberal administration can increase the oxygen SaO /SpO clearly above the patient's normal range, which may cause respiratory depression because of chemoreceptor insensitivity for CO2. Blood gases should be monitored to avoid excessive retention of CO2 in patients with hypercapnia or reduced CO2-sensitivity, in order to adjust the oxygen therapy.

Nebulisation
When used for nebulization, oxygen can be employed as the sole driving gas (100% oxygen at a sufficient flow rate to nebulize the fluid in the nebulization chamber) or mixed with air. In nebulization therapy, the flow rate of oxygen and/or oxygen mixed with air is typically a continuous flow of 6–8 liters per minute.

Fresh gas supply in anaesthesia or intensive care - Assisted or controlled ventilation
Oxygen is frequently used in intensive care settings and should be titrated to meet the individual patient's needs. It is typically administered through assisted or controlled ventilation.
Positive End-Expiratory Pressure (PEEP) is often applied to enhance ventilation/perfusion matching, recruit airways and lung volumes, and subsequently reduce shunting.
During general anesthesia, a fraction of inspired oxygen (FiO2) of about 0.3 is usually sufficient, although higher fractions may be necessary for some patients. When oxygen is mixed with other gases, the fraction of oxygen should be maintained at a minimum of 0.21 in the inhaled gas mixture, with the FiO2 potentially increased up to 1.0.

First aid treatment
In emergency situations requiring the acute administration of 100% oxygen, a face mask with a reservoir (ensuring sufficient oxygen flow to keep the reservoir from collapsing during inhalation) or a demand valve system should be utilized.

Cluster Headache
Oxygen administration should be instituted as soon as possible after the onset of the attack. Oxygen should be delivered by facemask in a non-re-breathing system with a continuous flow of 6 to 12 l/min, for about 15 min.

Paediatric population
The safety and efficacy of oxygen therapy are well established for children of all ages. Dosing for the paediatric population is the same as for adults, except for newborns (term, near-term, and preterm), where careful monitoring is essential. Oxygen concentrations of up to 100% may be used to ensure adequate oxygenation but should be administered for the shortest duration possible. During newborn resuscitation, guidelines recommend starting with air, seeking the lowest effective oxygen concentration to achieve adequate oxygenation. For initial therapy, oxygen at low concentrations (up to 40% FiO2) combined with CPAP is advised.
Oxygen therapy has no relevant use in the paediatric population for acute cluster headache, as it is not indicated for this condition in children.

Hyperbaric Oxygen Therapy (HBOT)
General recommendations
Qualified healthcare professionals should administer HBOT. HBOT means that 100% oxygen is delivered at a pressure higher than the atmospheric pressure at sea level (1 atmosphere=101.3 kPa=760 mmHg). For safety reasons the pressure of HBOT should not exceed 3 atmospheres. The duration of a single HBOT session at 2 to 3 atmospheres typically ranges from 60 minutes to 4–6 hours, depending on the indication. If needed, sessions can be repeated 2 to 3times per day based on the indication and the patient's clinical condition. Compression and decompression should be performed slowly, following standard protocols, to minimize the risk of barotrauma. The duration and frequency of treatment should be determined by the attending physician, considering the patient's physical condition and medications. Specific recommendations for each indication are outlined below.

Decompression sickness and air/gas embolism due to other reasons
HBOT at 2.5 to 3 atmospheres for up to 24 hours, repeated as needed, is recommended.

Osteo-radionecrosis and clostridial myonecrosis (gas gangrene)
For osteo-radionecrosis, 2.4 atmospheres for approximately 90 minutes is recommended while for clostridial myonecrosis 3 atmospheres for approximately 90 minutes is recommended. Treatment may be repeated based on the results of the therapy.

Carbon monoxide poisoning
2.5 to 3 atmospheres are recommended for HBOT. A typical treatment usually lasts 45 minutes.

Paediatric population

HBOT can be used in children of any age. The length and frequency of treatment should be decided by the attending physician taking the patient's physical and disease status into account.

h) Route of Administration

Inhalation.

i) Contraindications

Normobaric Oxygen Therapy: There is no formal contraindication to normobaric oxygen therapy.
Hyperbaric Oxygen Therapy: Contraindicated with undrained/untreated pneumothorax.

j) Warnings and Precautions

High oxygen concentrations should be given for the shortest possible time required to achieve the desired result, and must be monitored with repeated checks of arterial gas pressure (PaO) or haemoglobin oxygen peripheral saturation (SpO) and clinical assessment.

Oxygen can be administered safely in the following concentrations, for the periods indicated:
- Oxygen concentrations up to 100% for less than 6 hours.
- Oxygen concentrations of 60-70% for less than 24 hours.
- Oxygen concentrations of 40-50% during the second 24-hour period.

Oxygen is potentially toxic after two days in concentrations of > 40%.
Low oxygen concentrations should be used in patients with respiratory failure who depend on hypoxia as a breathing incentive. In such cases, careful monitoring of the treatment is required, by measuring the arterial oxygen pressure (PaO) or through pulsoxymetry (arterial oxygen saturation (SpO)) and clinical assessment.

Oxygen administration in patients with drug-induced respiratory failure (opioids, barbiturates) or with chronic obstructive pulmonary disease (COPD) could further aggravate respiratory failure due to hypercapnia caused by the high blood levels of carbon dioxide, which neutralises the effects of oxygen on receptors.

High concentrations of oxygen in air or in the inhaled air will cause the concentration and pressure of nitrogen to fall. This will also reduce nitrogen levels both in tissues and the lungs (alveoli). If oxygen is absorbed into the blood through the alveoli faster than it is supplied through ventilation, the alveoli may collapse (atelactasis). This may hinder the oxygenation of arterial blood, because no gas exchange takes place despite perfusion.
In patients with reduced sensitivity to carbon dioxide pressure in arterial blood, high oxygen levels can cause carbon dioxide retention. In extreme cases, this may lead to carbon dioxide narcosis.

Patients with risk of hypercapnic respiratory failure

Special precaution should be adopted in patients with low sensitivity to carbon dioxide in arterial blood or at risk of hypercapnic respiratory failure ("hypoxic training") (e.g. patients with chronic obstructive pulmonary disease (COPD), cystic fibrosis, pathological obesity, chest wall deformities, neuromuscular disorders, breathing depressants overdose). Supplemental oxygen administration can cause respiratory depression and increase of PACO
with subsequent symptomatic respiratory acidosis. In these patients, oxygen therapy should be carefully titrated. The target oxygen saturation to be reached can be lower than in other patients, and oxygen should be administered at a lower flow rate.

Special precautions in patients with bleomycin-induced pulmonary lesions

Pulmonary toxicity of high dose oxygen therapy can aggravate pulmonary lesions, even if administered several year after the initial lung lesion induced by bleomycine, and the target oxygen saturation level to be reached can be lower than in other patients.

Paediatric population

Due to the greater sensitivity of the neonate to supplemental oxygen, the lowest effective concentration of oxygen must be administered to obtain adequate oxygenation for neonates. Increase in PaO in preterm and term-born infants can cause retinopathy in prematurity, chronic pulmonary disease and intraventricular haemorrhage. It is recommended to start resuscitation in late-preterm or near-term infants with breathing room air instead with 100% oxygen. For preterm neonates, the optimal concentration of oxygen and the oxygen target are not precisely defined. When required, supplemental oxygen should be carefully monitored and guided through pulsoxymetry.

Hyperbaric oxygen

Administration of oxygen in the hyperbaric chamber should be carefully evaluated based on the benefit /risk balance, in the event of:
- Recurrent otitis and/or sinusitis, laryngocele, mastoid cavities, vestibular syndrome, loss of hearing and recent middle ear surgery
- Arterial hypertension not treated pharmacologically
- Ischaemic and/or congestive heart disease; in patients with acute coronary syndrome or acute myocardial infarction also requiring hyperbaric therapy, as well as in the case of carbon monoxide intoxication, hyperbaric therapy must be administered with caution due to the potential vasoconstriction associated with hyperoxia in coronary circulation
- Restrictive and/or highly restrictive lung diseases
- Severe anxiety, psychosis, claustrophobia
- Glaucoma, retinal detachment even after surgical treatment (compensation manoeuvres)
- Uncontrolled high fever
- History of seizures, epilepsy

Patients with diabetes mellitus
Hyperbaric oxygen may interfere with glucose metabolism. The vasoconstrictive effects of hyperbaric therapy can also impair subcutaneous absorption of insulin, making the patient hypoglycaemic. The physician may also consider monitoring blood glucose levels between hyperbaric therapy sessions.

Respiratory disorders
Due to decompression, at the end of the hyperbaric oxygen therapy, the volume of gas increases, while pressure in the chamber diminishes, and this can generate partial pneumothorax or aggravate an underlying pneumothorax. In a patient with a pneumothorax that has not been drained, decompression could cause tension pneumothorax to develop.
Moreover, considering the risk of expansion of the gas during the decompression phase of hyperbaric therapy, the benefit/risk balance of hyperbaric therapy should be carefully assessed in patients with insufficiently controlled asthma, pulmonary emphysema, chronic obstructive pulmonary disease (COPD), and recent chest surgery.

k) Interactions with Other Medicaments
Inhalation of high concentration of oxygen can exacerbate the pulmonary toxicity associated with drugs such as bleomycin (even if oxygen is given several years after the initial bleomycin-induced lung injury), nitrofurantoin, amiodarone and with paraquat intoxication. Supplemental oxygen should be avoided unless the patient is experiencing low oxygen levels (hypoxemia).

l) Pregnancy and Lactation
Pregnancy:

Normobaric Oxygen Therapy:
No known negative effects. When necessary, Oxygen can be used during pregnancy i.e. in case of vital indications, women either critically ill or with hypoxaemia.

Hyperbaric Oxygen Therapy:
In case of carbon monoxide intoxication in pregnant women, the use of hyperbaric oxygen has shown a benefit for the foetus although the documented experience with the use of hyperbaric oxygen in pregnant women is limited. In other situations, hyperbaric oxygen therapy should be used with caution in pregnancy as the impact on the foetus of a potential increase of oxidative stress from excess oxygen is unknown. The use of hyperbaric oxygen therapy should then be evaluated in each individual patient but is permissible in the case of vital indications during pregnancy.

Lactation:
No known negative effects on lactation. Oxygen therapy can be used during breastfeeding without risk to the infant. Hyperbaric oxygen therapy should be avoided during lactation.

m) Side Effects
Undesirable effects due to normobaric and hyperbaric oxygen therapy are tabulated in the table below.

Paediatric population
In the use of oxygen in premature neonates, retinopathy of prematurity (retrolental fibroplasia) may occur at concentrations greater than 40%.

Adverse reactions associated with Normobaric Oxygen Therapy (NBOT) and hyperbaric Oxygen Therapy (HBOT):

	Very common (>1/10)	**Common (≥1/100 to <1/10)**	**Uncommon (≥1/1,1000 to <1/100)**	**Rare (≥10,000 to <1/1,000)**	**Very rare (≤1/10,000)**	**Not Known**
Respiratory, thoracic and mediastinal disorders				<u>**HBOT**</u> Dyspnoea		<u>**NBOT**</u> Pulmonary toxicity: • Tracheobronchitis • (substernal pain, dry cough) • Interstitial oedema • Pulmonary fibrosis Worsening of hypercapnia in patients with chronic hypoxia/hypercapnia treated with too much elevated FiO2: • Hypoventilation • Respiratory acidosis • Respiratory Arrest <u>**HBOT**</u> Respiratory disturbances
Nervous system disorders		<u>**HBOT**</u> Seizure				
Musculoskeletal and connective tissue disorders						<u>**HBOT**</u> Localised muscular twitching.
Ear and labyrinth disorders	<u>**HBOT**</u> Ear pain		<u>**HBOT**</u> Tympanic membrane rupture			<u>**HBOT**</u> • Vertigo • Hearing impaired Acute serous otitis media • Tinnitus
Gastrointestinal disorders						<u>**HBOT**</u> Nausea
Psychiatric disorders						<u>**HBOT**</u> Abnormal behaviour
Eye disorders	<u>**NBOT**</u> Retinopathy of prematurity <u>**HBOT**</u> Progressive myopia					<u>**HBOT**</u> • Peripheral vision decreased • Blurred vision cataract *

Injury, poisoning and procedural complications	**<u>HBOT</u>** Barotrauma (sinuses, ear, lung, teeth etc.)					
Metabolism and nutrition disorders				**<u>HBOT</u>** Hypoglycemia in diabetic patients.		
General disorders and administration site conditions						**<u>NBOT</u>** Mucosal dryness. Local irritation and inflammation of the mucosa.

n) Symptoms and Treatment of Overdose

Normobaric Oxygen Therapy:

The symptoms of oxygen intoxication are those of hyperoxia. The symptoms of respiratory toxicity include from tracheobronchitis (dry cough, substernal pain) to interstitial oedema and pulmonary fibrosis.

In case of oxygen intoxication related to hyperoxia, oxygen therapy should be reduced or if possible stopped, and symptomatic treatment should be started in order to maintain vital functions (e.g. artificial ventilation/assisted ventilation should be given if the patient shows signs of failing respiration).

Additional information on patients at risk of hypercapnic respiratory failure

The administration of supplemental oxygen may cause respiratory depression and a rise in partial pressure of Carbon Dioxide with subsequent symptomatic respiratory acidosis.

Paediatric population

Eye toxicity in neonates: in premature neonates who have been subjected to high oxygen concentrations, retinopathy of prematurity may occur.

Hyperbaric Oxygen Therapy:

In hyperbaric oxygen therapy, symptoms of central nervous toxicity such as tinnitus, respiratory disturbances, localized muscular twitching especially eyes, mouth, forehead are observed. Continuation of exposure can lead to vertigo and nausea followed by altered behaviour (confusion, anxiety, irritability), and finally generalized convulsions. Besides, eye toxicity includes blurred vision and reduced peripheral vision are also observed.

o) Effects on Ability to Drive and Use Machine

Under normal conditions, oxygen does not interfere with consciousness, however patients who require continuous oxygen support will require individual assessment, taking their entire medical situation into account for evaluating their ability to drive or operate machinery.

p) **Instruction for Use**

Make sure the cylinder is well secured before using.
Must be fitted with MDA approved regulator (pressure-reducing device).
Slowly open the cylinder valve.
Adjust the regulator to desired flow.
Ensure to close the cylinder valve when not in use

q) Storage Conditions

Medicinal Oxygen cylinders should be:
- Prohibiting smoking and naked flame in the storage area.
- Cylinders should be stored indoor, kept dry and clean.
- Kept away from any flammable materials.
- Store in a well-ventilated place.
- Store in upright position and secure it.

- Full and used cylinders should be stored separately. Full cylinders should be used in strict rotation.
- Medical cylinders containing different gases should be segregated and identified within the store.
- Cylinders must not be repainted, have any markings obscured or labels removed.
- Store at temperature below 50°C.

r) Dosage forms and packaging available

Dosage: Medicinal gases, compressed.

Packaging:

Cylinder Material	Cylinder Size (L)	Valve	Pressure (bar)	Content (m^3)
Carbon Steel or Aluminium	2.3	Bullnose (BS 341) or Pin Index (ISO 407 or CGA870)	150	0.35
	3.2-3.4		150	0.5
	5		150	0.7
	10		150	1.4
	35		150	5.1
	45		150	6.8
	46.7		150	7.0
	47		150	7.2
	50		150	8.0
	50		200	10.7

s) Name and address of manufacturer/ product registration holder

Mega Mount Industrial Gases Sdn. Bhd, (451911-V)

PLO 192, Jalan Siber 8, Kawasan Perindustrian Senai IV, 81400 Senai, Johor, Malaysia.

t) Product Registration Number

MALxxxxxxX

u) Date of revision of PI

Rev.01 17-Mar-2025